IE University
Master in Business Analytics & Data Science

Master's Thesis

Mean, Variance, and Market Efficiency in the Silver Market

Forecasting Weekly Returns and Realised Volatility with Classical, Machine-Learning, and Deep-Learning Models

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Abstract

This thesis asks how much of silver's short-term price behaviour can be forecast, and frames the question through the efficient-market hypothesis (EMH). Silver is modelled at the weekly (Friday-to-Friday) horizon over 2015–2026, and the analysis is factorised into the two moments: the conditional *mean* of returns and the conditional *variance*—efficiency constrains the former but not the latter. For the mean, a deliberate ladder of models of increasing flexibility—ARIMA/ARIMAX, VAR, mixed-data-sampling (MIDAS) regressions, random forests, gradient boosting, and an LSTM—is tested against a common random-walk-with-drift benchmark, expanding the information set from silver's own past (weak-form EMH) to a broad set of public information (semi-strong-form EMH): cross-asset returns, silver futures positioning, macroeconomic releases, technical indicators, and sentiment from Reddit and financial news. For the variance, weekly realised volatility is modelled with GARCH and the heterogeneous autoregressive (HAR) model, and the question of interest is whether sentiment improves the forecast. The central finding is a single coherent statement: no model beats the drift benchmark for the conditional mean by a significant margin—weak- and semi-strong-form efficiency hold for weekly silver returns—yet the conditional variance is strongly and significantly forecastable, with sentiment adding a small but robust increment. Predictable volatility alongside an unpredictable mean is exactly what the EMH permits, so the two results are complementary rather than contradictory.

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1 Introduction

Silver lives a double life. It is an industrial input—roughly half of annual demand goes into photovoltaics, electronics, brazing alloys, and medical uses, tying its price to the global manufacturing cycle—and, at the same time, a monetary metal held as a store of value, an inflation hedge, and a more volatile cousin of gold. On top of both sits a retail following that can sometimes turn silver into a meme trade: in late January 2021, Reddit's r/WallStreetSilver drove spot from roughly $25 to $30 an ounce within days before it faded. This blend of industrial demand, monetary hedging, and episodic retail attention makes silver an unusually rich laboratory for the oldest question in empirical finance: how much of a price can be forecast?

That question has a canonical answer, and it is mostly negative. The efficient-market hypothesis (Fama, 1970, 1991) holds that prices already embed available information, so successive price changes are approximately unpredictable. It is conventionally stated in three nested forms, graded by the information set a forecaster is allowed to use:

- **Weak form** — past prices and returns carry no exploitable signal.
- **Semi-strong form** — no *public* information does either: not prices, fundamentals, news, or macroeconomic data.
- **Strong form** — not even *private*, inside information does.

Each form nests the one before it, so the claims grow progressively stronger. For a liquid, globally traded metal, weak- and semi-strong-form efficiency are the natural hypotheses to take to the data, and they are the two this thesis tests.

Empirically, silver's *price* behaves like a random walk with drift: it is non-stationary, and shocks accumulate rather than mean-revert (Figure 1). Returns, however, are stationary—they fluctuate around a stable long-run mean rather than wandering without bound. The object worth modelling is therefore not the price level but its increment, the weekly log-return $r_t = \log P_t - \log P_{t-1}$; stationarity is what makes the forecasting problem well-posed, giving a fixed quantity to predict and a natural yardstick—that same prevailing average—to judge any forecast against.

The weekly (Friday-to-Friday) horizon is a deliberate choice of frequency, made where the information aligns. Futures-market positioning is reported weekly and macroeconomic releases monthly, while daily sentiment is too sparse and noisy to inform a forecast and daily returns are dominated by microstructure noise; a weekly horizon is coarse enough to average out that noise yet fine enough to leave roughly six hundred observations over 2015–2026—enough to train and properly out-of-sample-test the machine-learning models. Weekly realised volatility, the target of the second-moment analysis, is in turn aggregated from daily returns.

Figure 1: Silver front-month futures (SI=F), daily, January 2015–May 2026. The annotated episodes punctuate the sample: the COVID-19 crash (March 2020), the r/WallStreetSilver squeeze (January 2021), the start of the Federal Reserve's hiking cycle (March 2022), and the 2024–2026 bull run that carried spot to an all-time high near \$115/oz before a sharp (-41%) correction. The level is non-stationary and shocks accumulate; the thesis models its stationary weekly log-return rather than the level itself.

The efficiency question then becomes whether r_t can be forecast—and silver's own past offers little to go on. Weekly returns carry almost no linear autocorrelation, so a linear own-history model such as ARIMA has little to grip; a formal model search duly collapses it to a constant mean. Yet the returns are not independent noise either, because their *magnitude* is persistent—volatile weeks cluster together—so structure remains, only in the variance rather than the mean. The thesis follows this thread: it asks first whether silver's own history forecasts the return, then whether the broad public-information set a real forecaster would possess does—cross-asset returns (gold, the US dollar, copper, equities, oil, and the VIX), futures-market positioning, macroeconomic releases, technical indicators, and retail and news sentiment—and, separately, whether the predictable variance can be modelled.

Research questions. The thesis is built around four questions:

- **Q1 (weak form).** Are weekly silver returns forecastable from their own past?
- **Q2 (semi-strong form).** Does expanding the information set to public covariates help—and does it matter whether the relationship is assumed linear or allowed to be nonlinear?
- **Q3 (the second moment).** Is weekly realised volatility forecastable, and if so by how much?
- **Q4 (sentiment).** Does retail and news sentiment add incremental predictive

content to either the conditional mean or the conditional variance?

Approach. To answer these questions, the analysis works at the weekly (Friday-to-Friday) horizon and estimates a deliberate ladder of models of increasing flexibility against a common benchmark and a unified evaluation protocol. The benchmark throughout is the prevailing-mean "drift" forecast—the random-walk-with-drift, equivalent to ARIMA(0,0,0)—which is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008). For the conditional mean, the ladder runs from classical time-series models (ARIMA/ARIMAX, VAR, and MIDAS regressions) through tree ensembles (random forests and gradient boosting) to a recurrent neural network (LSTM); for the conditional variance it runs GARCH and the heterogeneous autoregressive (HAR) model of Corsi (2009) alongside the same tree ensembles. Forecasts are compared on out-of-sample R^2 relative to the drift (Campbell & Thompson, 2008), the Diebold–Mariano test of equal predictive accuracy (Diebold & Mariano, 1995), and—for the directional reading—the Pesaran–Timmermann test (Pesaran & Timmermann, 1992), all on a held-out 2023–2026 test window.

Findings in brief. The results tell one coherent story. No model—linear or nonlinear, own-history or public-information—beats the drift benchmark for the conditional mean by a statistically significant margin; feature-rich and shorter-window variants do significantly worse. Weak- and semi-strong-form efficiency hold for weekly silver returns, and they hold more cleanly once the exceptional 2025 bull run is set aside. The conditional variance is a different matter: HAR and GARCH both forecast weekly realised volatility well clear of the naïve benchmark, and Reddit attention adds a small but statistically robust increment on top. Crucially, predictable volatility does *not* contradict efficiency—it is precisely the mean-versus-variance distinction drawn above—so the two halves of the thesis are not a success and a failure but two views of the same result: *weekly silver returns are unforecastable in the mean from past prices and public information, yet their variance is strongly predictable.* A weak, non-replicating directional signal from futures-market positioning is the single caveat, and it is best read through the adaptive-markets lens of Lo (2004) rather than as evidence against efficiency.

Contributions. The thesis makes four contributions. First, it evaluates a full model ladder—classical, machine-learning, and deep-learning—for *both* moments of weekly silver returns under one drift benchmark and one significance battery, so that model families are compared on a level footing rather than across incommensurable setups. Second, it makes the conditional-mean / conditional-variance factorisation explicit, turning what could read as "returns failed, volatility worked" into a single coherent efficiency statement. Third, it builds a domain-appropriate sentiment pipeline—Twitter-

RoBERTa for Reddit, FinBERT for financial news—and treats sentiment as a falsifiable ablation on both moments rather than assuming it helps. Fourth, it documents a carefully qualified set of negative results on semi-strong predictability of the mean, separating magnitude from direction and showing where an apparent directional signal fails to replicate across model classes.

Structure. The remainder of the thesis is organised as follows. Section 2 sets out the conceptual framework—the random-walk view of prices, the martingale-difference characterisation of the conditional mean, the expanding information set, and the linear-versus-nonlinear alternatives that motivate the model ladder. Section 3 reviews the relevant literature on commodity-return predictability, mixed-frequency modelling, realised-volatility forecasting, and financial sentiment. Section 4 describes the data and feature construction. Section 5 presents the models and the unified evaluation protocol. Section 6 reports the conditional-mean (returns) results, and Section 7 the conditional-variance (volatility) results. Section 8 discusses the findings against the efficiency framing, and Section 9 concludes with limitations and directions for future work.

2 Conceptual Framework

This section fixes the notation used throughout and states, precisely, the nulls that each model class is built to test. The aim is to keep the language exact without overclaiming: the empirical results are framed as tests of the conditional mean and the conditional variance of weekly silver returns, not as claims about the full return distribution.

2.1 Prices, returns, and the random walk

Let P_t be the silver price and $p_t = \log P_t$ the log price. The weekly log-return is the increment in the log price,

$$r_{t+1} = p_{t+1} - p_t, \qquad \text{equivalently} \qquad p_{t+1} = p_t + r_{t+1}, \tag{1}$$

so the price level and the return are two views of the same process: the level accumulates shocks, while returns *are* the shocks. A random walk with drift writes the level as $p_{t+1} = p_t + \mu + \varepsilon_{t+1}$ with $\mathbb{E}[\varepsilon_{t+1}] = 0$. Silver log prices are best described as *random-walk-like*—non-stationary, with shocks that accumulate—which is the reason the modelling target throughout is the stationary increment r_t rather than the level. The safer wording avoids the *strict* random walk: the innovations are not independent and identically distributed, because their variance clusters (Section 2.4).

2.2 The conditional mean and the information set

Let F_t denote the information available at time t. For the weak-form test this is silver's own past, $F_t = \{r_t, r_{t-1}, r_{t-2}, \ldots\}$. Returns are a *martingale difference sequence* (MDS) with respect to F_t if the conditional mean equals a constant drift,

$$\mathbb{E}[r_{t+1} \mid F_t] = \mu, \tag{2}$$

that is, past returns do not move the expected next-week return away from its long-run average. This is a statement about the first moment *only*; it does not assert that returns are i.i.d. The semi-strong test expands the information set to public covariates,

$$G_t = \{F_t,\ \text{gold},\ \text{USD},\ \text{copper},\ \text{VIX},\ \text{oil},\ \text{COT},\ \text{macro},\ \text{technicals},\ \text{sentiment}\}, \tag{3}$$

and asks whether the martingale-difference property survives the larger conditioning set,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu. \tag{4}$$

ARIMA tests whether silver returns are a martingale difference with respect to their own past, Equation (2); ARIMAX and the machine-learning models test whether that property survives once the information set is expanded to public covariates, Equation (4). Operationally, both nulls are tested as "does the model beat the prevailing-mean (drift) benchmark out of sample?"—a forecast that beats the drift would contradict the corresponding null.

2.3 Linear and nonlinear alternatives

The alternative to Equation (4) can be specified narrowly or broadly. A linear model entertains an additive alternative,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu + \beta' G_t, \tag{5}$$

so ARIMAX, VAR, and MIDAS ask whether public covariates enter the conditional mean linearly and additively. Nonlinear models entertain the broader alternative

$$\mathbb{E}[r_{t+1} \mid G_t] = f(G_t), \tag{6}$$

where f may encode thresholds, interactions, regime dependence, and sequence effects. The motivation is concrete: a signal may be invisible to any single covariate yet present in a combination. Each condition alone may carry little signal, $\mathbb{E}[r_{t+1} \mid \text{high sentiment}] \approx \mu$

and $\mathbb{E}[r_{t+1} \mid \text{stretched COT}] \approx \mu$, while the joint condition does not,

$$\mathbb{E}[r_{t+1} \mid \text{high sentiment, stretched COT, rising VIX, falling USD}] \neq \mu. \tag{7}$$

Random forests, gradient boosting, and the LSTM search this larger space. The logic of the ladder is therefore cumulative: if even flexible nonlinear models fail to beat the drift benchmark out of sample, the martingale-difference interpretation is *strengthened* rather than merely assumed from linear tests. A complementary own-history check, the generalised spectral test (Hong & Lee, 2003), probes for nonlinear departures from Equation (2) that linear autocorrelation diagnostics cannot see.

2.4 The conditional variance

Efficiency constrains Equation (4); it places no restriction on the conditional variance $\text{Var}(r_{t+1} \mid F_t)$. The two can diverge: the mean may be (approximately) constant while the variance moves through time, which is exactly what volatility clustering means—a large move today says little about the sign or size of next week's return but does signal that next week is likely to be more volatile. Strict white noise would require independence in r_t, r_t^2, and $|r_t|$ alike; the MDS property in Equation (2) restricts only the first of these. Weekly silver returns are close to white noise in the mean but not strict white noise, because squared returns are strongly autocorrelated. The predictable structure is therefore concentrated in the second moment, which is what the volatility chapter (Section 7) targets directly via realised volatility. Figure 2 shows both moments at once for the weekly silver return.

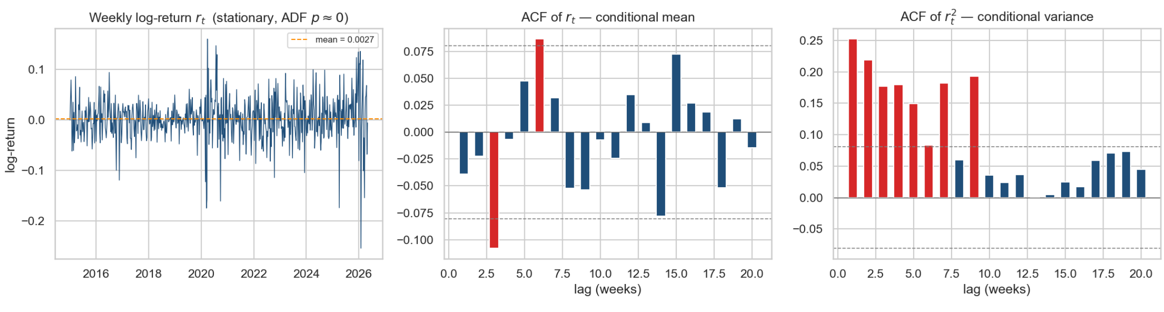

Figure 2: The two moments of the weekly silver log-return, 2015–2026. *Left:* the return series—stationary, fluctuating around a small positive drift (mean $\approx$ 0.0027/week). *Centre:* the sample autocorrelation of returns lies almost entirely within the 95% white-noise band (dashed), with only isolated lags breaching it—consistent with the martingale-difference null of Equation (2), so a linear own-history model has little to exploit. *Right:* the autocorrelation of *squared* returns is large and positive across the first several lags—the volatility clustering of Section 2.4. The mean is close to unpredictable; the variance is not.

2.5 Direction versus magnitude

A final distinction governs how forecasts are scored. Directional accuracy asks whether $\text{sign}(\hat{r}_{t+1}) = \text{sign}(r_{t+1})$ more often than chance, which is *not* the same as beating the drift benchmark under squared-error loss: a model can call the sign correctly yet fail to improve out-of-sample R^2. Because efficiency is a conditional-mean claim, the magnitude axis (RMSE, out-of-sample R^2, and Diebold–Mariano against the drift) is load-bearing, while directional accuracy—and its significance test, Pesaran–Timmermann (Pesaran & Timmermann, 1992)—is a secondary, economically motivated lens. Directional gains are interesting but do not by themselves reject the martingale-difference null.

3 Literature Review

The thesis sits at the intersection of five strands of literature: the long empirical record on commodity-return predictability and market efficiency, mixed-frequency regression, machine learning for financial forecasting, realised-volatility modelling, and text-based sentiment. This section reviews each in turn, with an eye on the specific design choices they motivate in the chapters that follow.

3.1 Commodity-return predictability and market efficiency

The efficiency framing of Section 2 descends directly from Samuelson (1965) and the weak/semi-strong/strong taxonomy of Fama (1970), whose later survey (Fama, 1991) reframed the weak form more broadly as "tests for return predictability"—precisely the form the present exercise takes. For commodities specifically, the canonical stylised facts are discouraging for forecasters: Deaton and Laroque (1992) document spot prices that are extremely volatile, dominated by long quiet spells punctuated by sharp spikes, and difficult to distinguish from highly persistent, near-unit-root processes—which is why the stationary increment (the return), not the level, is the only defensible modelling target. At the same time, commodities do not move in isolation: Zhang and Wei (2010) find cointegration and price discovery between crude oil and gold, and Batten et al. (2010) show that the macroeconomic determinants of volatility differ across the precious metals, with silver responding to a different mix of monetary variables than gold. Together these findings motivate the cross-asset information set used here—gold, the dollar, copper, equities, the VIX, and oil—while cautioning against treating silver as a small gold.

Whether any of that public information translates into out-of-sample forecasting power is the subject of a debate the thesis borrows its evaluation protocol from. Welch and Goyal (2008) re-examine the full catalogue of variables claimed to predict the equity premium and find that essentially none beats the prevailing historical mean once evaluated out of sample; Campbell and Thompson (2008) reply that modest gains are possible, but only with theory-imposed restrictions, and propose the out-of-sample R^2 against the prevailing-mean forecast as the natural yardstick. That benchmark—the random-walk-with-drift, estimated as an expanding historical mean—and that metric are adopted unchanged in Section 5. Finally, Lo (2004) offers the adaptive-markets reading in which efficiency is not a binary property but an evolutionary, time-varying one, so that pockets of predictability can appear and decay; this is the natural frame for the episodic, non-replicating directional signals encountered in Section 6.

3.2 Mixed-frequency models (MIDAS)

Forecasting a weekly return with monthly macroeconomic releases and daily market data poses an alignment problem: aggregating the high-frequency series throws information away, while including every high-frequency lag as a free regressor invites massive parameter proliferation. Mixed-data-sampling (MIDAS) regression (Ghysels et al., 2004, 2007) resolves the trade-off by letting the high-frequency lags enter through a low-dimensional weight polynomial—typically the Beta or exponential-Almon family—so that a long lag window costs two or three parameters instead of dozens. Andreou et al. (2013) demonstrate the payoff in the direction most studied, using daily financial data to forecast quarterly macro aggregates; the present application inverts that direction, bringing monthly macro releases and daily cross-asset returns to bear on a weekly financial target. The `midasr` package (Ghysels et al., 2016) documents the now-standard estimation approach—nonlinear least squares over the weight parameters with the linear coefficients profiled out—which the Python implementation in this thesis follows.

3.3 Machine learning and deep learning for financial time series

Tree ensembles and recurrent neural networks instantiate the nonlinear alternative $f(G_t)$ of Section 2.3: random forests and gradient boosting capture thresholds and interactions without specifying them in advance, and the long short-term memory network (Hochreiter & Schmidhuber, 1997) adds explicit sequence memory, extracting its own representation of the recent past rather than consuming hand-picked lags. The applied literature is large and, on its face, encouraging—the systematic review of Sezer et al. (2020) catalogues hundreds of deep-learning forecasting studies reporting gains over classical models. It is also methodologically uneven: benchmarks are often weak or absent, evaluation windows short, and statistical significance rarely assessed, so reported out-performance frequently dissolves against an honest no-change or prevailing-mean benchmark. The design response in this thesis is to keep the model ladder flexible but the referee strict: every learner, however expressive, is scored against the same drift benchmark with the same Diebold–Mariano test, so that flexibility can only ever strengthen—never substitute for—the efficiency verdict.

3.4 Realised volatility and the HAR model

That the conditional variance, unlike the conditional mean, is forecastable has been textbook knowledge since Engle (1982) introduced ARCH and Bollerslev (1986) generalised it: volatility clusters, so its own past predicts it. The realised-volatility literature sharpened the target, replacing latent variance with a model-free estimate built from higher-frequency squared returns—here, weekly realised volatility from daily returns. Its workhorse forecaster is the heterogeneous autoregressive (HAR) model of Corsi (2009), which regresses next period's realised volatility on its own past averaged over short, medium, and long horizons; despite being estimable by OLS, the cascade of horizons approximates the long memory of volatility well enough that HAR remains the standard benchmark. Evaluation has its own subtlety: realised volatility is itself a noisy proxy of the latent target, and Patton (2011) shows that only a restricted family of loss functions—in practice MSE and QLIKE—ranks forecasts consistently under proxy noise, with QLIKE the more powerful in heavy-tailed samples. The volatility chapter therefore reports QLIKE-based Diebold–Mariano tests as primary. None of this strains the efficiency hypothesis, which constrains the mean and not the variance (Section 2.4): predictable volatility is permitted, and indeed expected.

3.5 Financial sentiment and investor attention

That media tone carries information for markets predates modern language models: Tetlock (2007), using a simple dictionary count of a Wall Street Journal column, found that media pessimism predicts downward pressure on prices followed by reversal. What has changed is measurement. Transformer language models (Devlin et al., 2019; Liu et al., 2019) replaced dictionaries with contextual representations, and domain-adapted variants—FinBERT for financial news (Araci, 2019) and Twitter-RoBERTa for social-media text (Loureiro et al., 2022)—now score sentiment with near-annotator accuracy on their home domains; both are used here, each on the text type it was trained for. On where such signals should help, the evidence points to the second moment: Audrino et al. (2020) find that sentiment and attention measures add significant predictive power for stock-market volatility—with attention the stronger of the two—but only marginal power for returns, and Smales (2021) documents investor attention, proxied by Google searches, as a state variable for market reactions under stress. Silver brings a natural laboratory for these attention effects: the January 2021 r/WallStreetSilver episode (Section 1) was an explicitly coordinated retail-attention shock, and the Reddit and news series collected here cover it. The volatility chapter's sentiment ablation is designed exactly along this literature's fault line: attention versus tone, variance versus mean.

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